

Problem 2.10

For the interval $0 \leq t \leq 10$, determine the accumulation function $a(t)$ that corresponds to

$$A(t) = 225 - (t - 10)^2.$$

Solution

The accumulation function is defined by

$$a(t) = \frac{A(t)}{A(0)}.$$

First compute $A(0)$:

$$A(0) = 225 - (0 - 10)^2 = 225 - 100 = 125.$$

Therefore,

$$a(t) = \frac{225 - (t - 10)^2}{125}.$$

Hence, the required accumulation function is

$$\boxed{a(t) = \frac{225 - (t - 10)^2}{125}}, \quad 0 \leq t \leq 10.$$